LA Lab - 4

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Aim : To compare the performance and features of different file systems (ext2, ext3, ext4, NTFS, FAT32) using benchmarking tools.

Theory :

A file system acts as an abstraction layer between the operating system and physical storage, defining how data is stored, organized, accessed, and protected.

Modern file systems go beyond basic storage and incorporate advanced mechanisms such as:

Journaling

- Maintains a log of operations before applying them
- Ensures data consistency and crash recovery
- Used in: ext3, ext4, NTFS

Block Allocation Strategies

- Determines how disk blocks are assigned to files
- Aims to reduce fragmentation and improve performance

Metadata Management

- Stores file information such as:
  - Permissions
  - Ownership
  - Timestamps
- Managed via:
  - Inodes (Linux: ext2/3/4)
  - MFT (Master File Table) (NTFS)

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2. **Architectural Characteristics of File Systems**

ext2 (Second Extended File System)

- Non-journaling → no logging overhead
- High performance for simple workloads
- Major drawback:
  - Requires full fsck (file system check) after crash
- Suitable for:
  - Flash drives
  - Low-resource systems

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ext3 (Third Extended File System)

- Adds journaling to ext2
- Supports atomic updates

- Slower than ext2 due to logging overhead
- Uses indirect block mapping → less efficient for large files

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ext4 (Fourth Extended File System)

- Advanced version of ext3
- Uses extents (contiguous block allocation) instead of block lists

Advantages:

- Reduced fragmentation
- Faster large file handling
- Better scalability (multi-terabyte volumes)
- Optimized journaling

Result: Best overall performance + reliability

NTFS (New Technology File System)

- Developed by Microsoft
- Feature-rich and complex

Key Features:

- Journaling
- ACL (Access Control Lists)
- Encryption (EFS)
- Sparse files

Designed for:

- Enterprise systems
- Windows OS environments

FAT32 (File Allocation Table)

- Very simple structure
- No journaling, no security

Limitations:

- Max file size = 4GB
- High fragmentation

Advantage:

- Extremely low overhead
- Works across all operating systems

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3. **Analysis of Benchmark Results**

A. Sequential Performance (SEQ1M)

Sequential Read

- NTFS (6182 MB/s) → Best performance
- Reason:
    - Windows optimized for large file streaming
    - Efficient read-ahead mechanisms

- ext family:
    - ext4 performs best among Linux systems due to extents

Sequential Write

- ext4 (3873 MB/s) → Highest performance
- NTFS shows a large drop (1086 MB/s)

Reason:

- ext4:
    - Uses extent-based allocation
    - Efficient write buffering
- NTFS:
    - More metadata operations → overhead

B. Random Performance (RND4K)

Random Read

- ext4 (1872 MB/s) significantly outperforms NTFS (560 MB/s)

Reason:

- Linux I/O scheduler handles:
    - Multiple parallel requests better
- ext4:
    - Efficient indexing and reduced fragmentation

Random Write

- ext2 and ext4 ≈ 429 MB/s
- NTFS and FAT32 perform worse

Important Insight:

- ext2 (no journaling) ≈ ext4 (with journaling)

Meaning:

- ext4 journaling is highly optimized
- Journaling overhead is minimal in modern systems

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4. FAT32 as a Special Case

FAT32 shows:

- Very high sequential write (3635 MB/s)

Reason:

- No journaling
- Minimal metadata operations
- Direct data writing

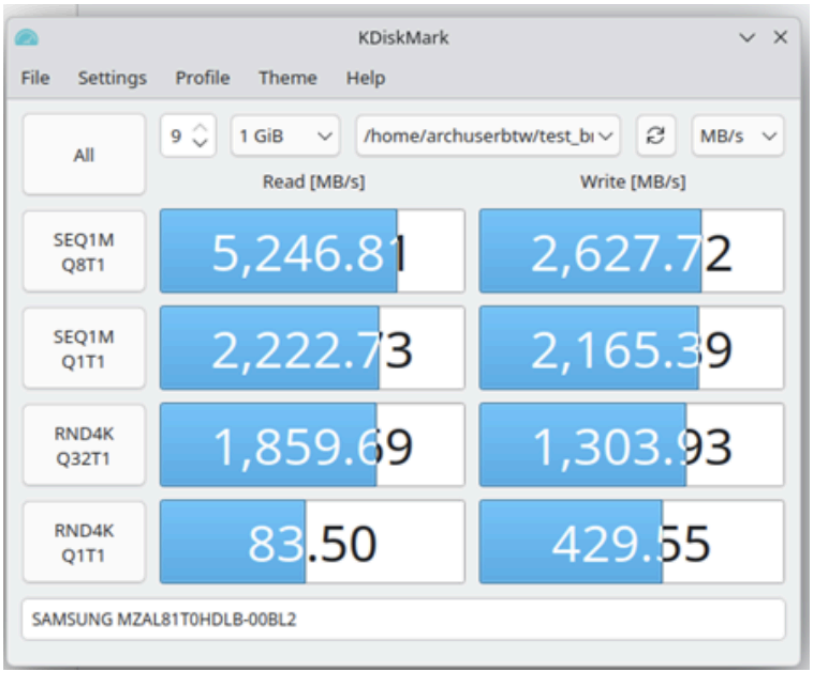
Trade-off:

- No reliability
- No security
- Higher risk of corruption

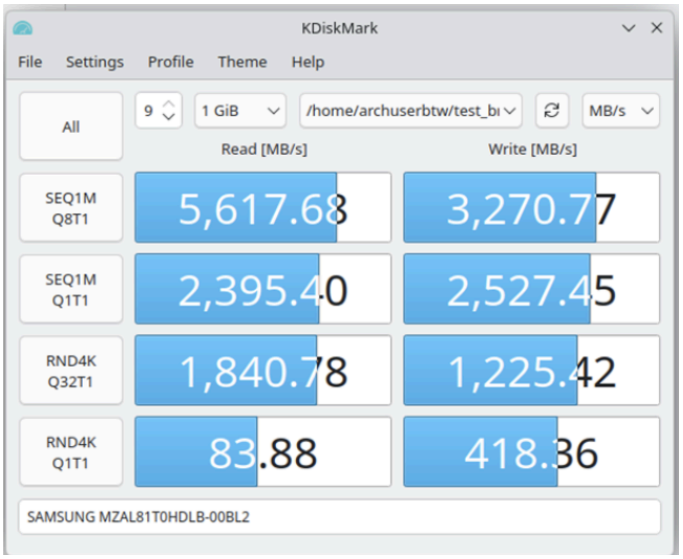
Feature	ext2	ext3	ext4	NTFS	FAT32
Journaling	❌	✅	✅	✅	❌
Max File Size	~2TB	~2TB	~16TB	~16TB	4GB
Performance	Fast	Medium	Fastest	Medium	Fast (small files)
Reliability	Low	Medium	High	High	Low
Fragmentation	Low	Low	Very Low	Medium	High
OS Support	Linux	Linux	Linux	Windows/Linux	All OS

Output :

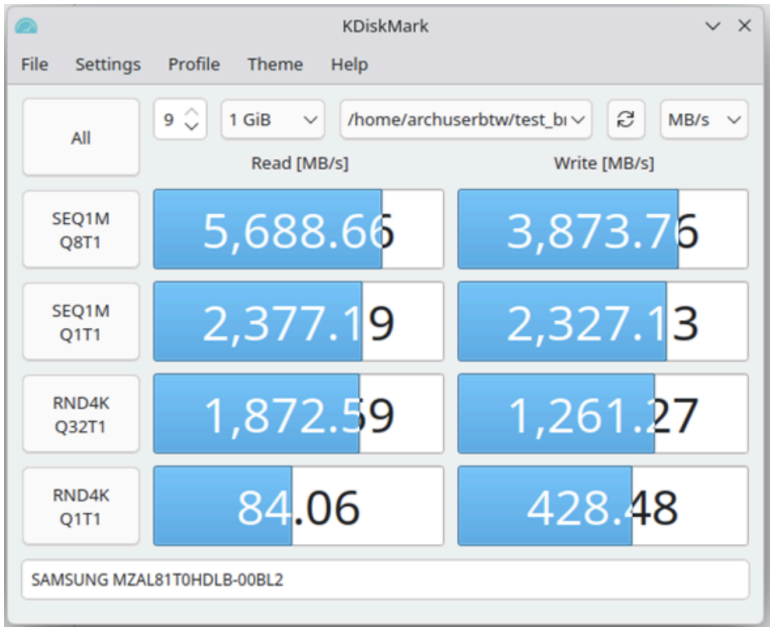
ext2



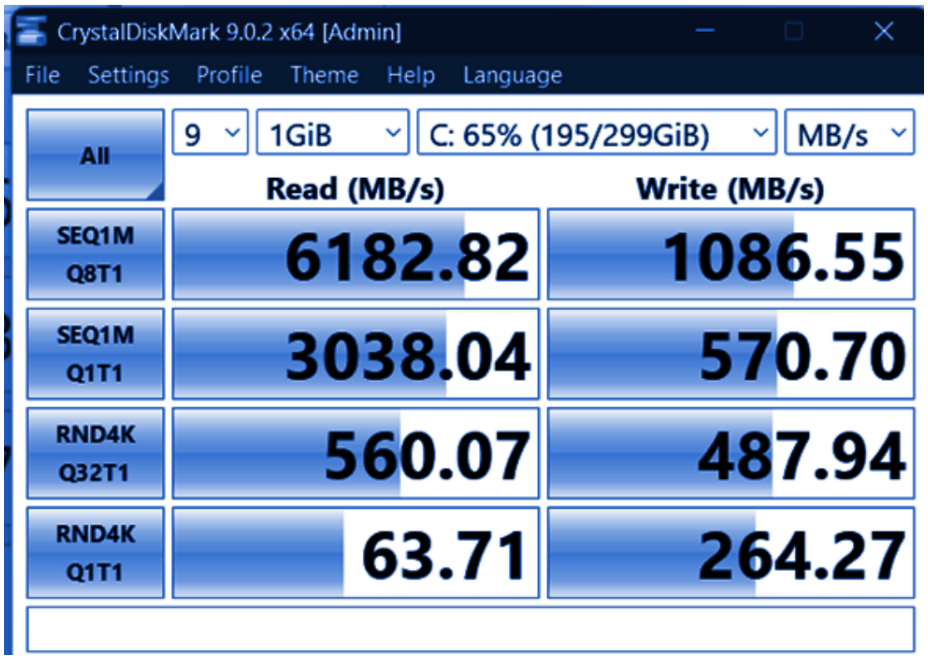
ext3


KDiskMark
File
Settings
Profile
Theme
Help
All
9
1 GiB
/home/archuserbtw/test_bi
MB/s
Read [MB/s]
Write [MB/s]
SEQ1M Q8T1
5,617.68
3,270.77
SEQ1M Q1T1
2,395.40
2,527.45
RND4K Q32T1
1,840.78
1,225.42
RND4K Q1T1
83.88
418.36
SAMSUNG MZAL81T0HDLB-00BL2

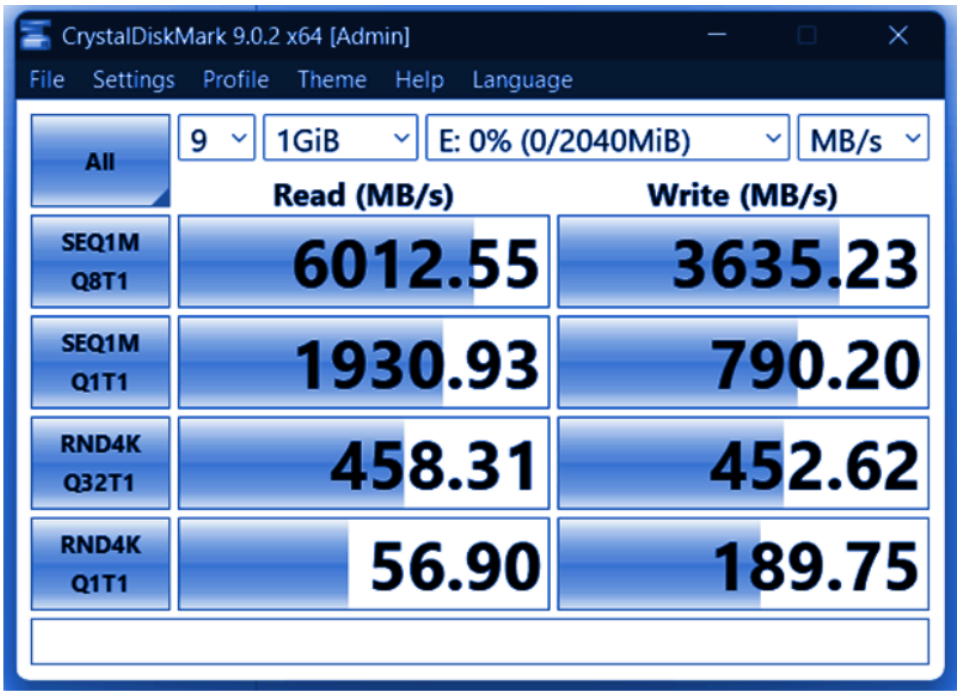
ext4

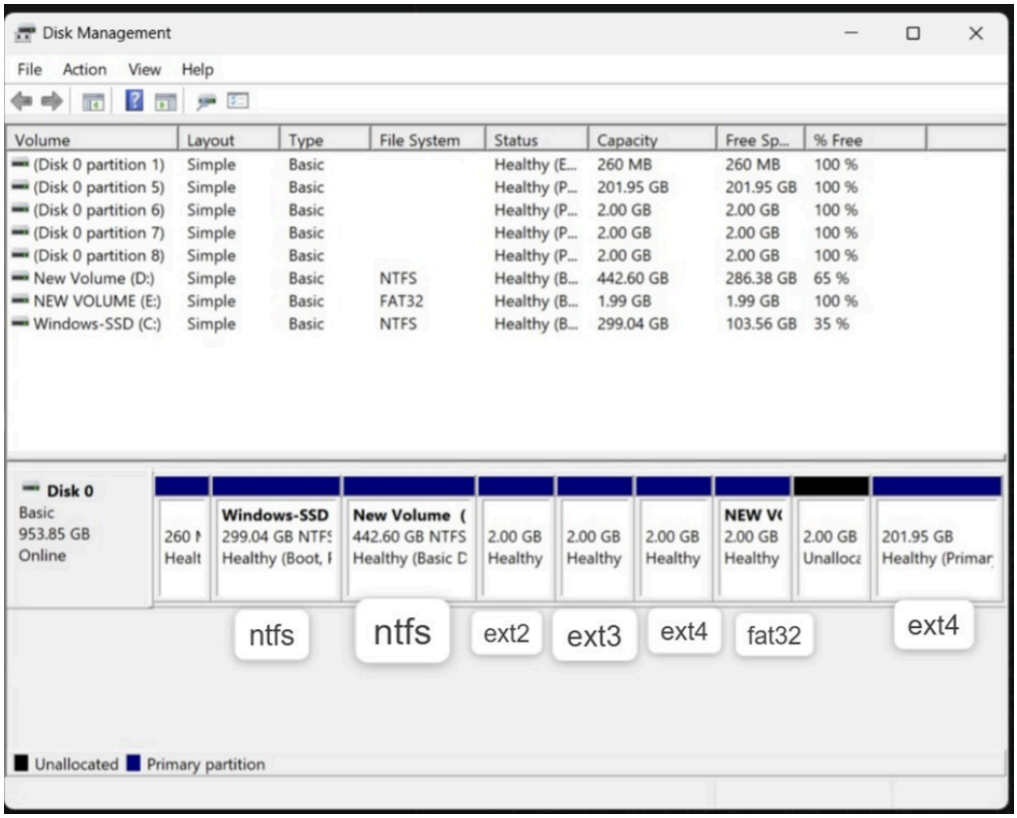

KDiskMark
File
Settings
Profile
Theme
Help
All
9
1 GiB
/home/archuserbtw/test_bi
MB/s
Read [MB/s]
Write [MB/s]
SEQ1M Q8T1
5,688.66
3,873.76
SEQ1M Q1T1
2,377.19
2,327.13
RND4K Q32T1
1,872.59
1,261.27
RND4K Q1T1
84.06
428.48
SAMSUNG MZAL81T0HDLB-00BL2

NTFS


CrystalDiskMark 9.0.2 x64 [Admin]
File Settings Profile Theme Help Language
All
9
1GiB
C: 65% (195/299GiB)
MB/s
Read (MB/s)
Write (MB/s)
SEQ1M Q8T1
6182.82
1086.55
SEQ1M Q1T1
3038.04
570.70
RND4K Q32T1
560.07
487.94
RND4K Q1T1
63.71
264.27

FAT32


CrystalDiskMark 9.0.2 x64 [Admin]
File Settings Profile Theme Help Language
All
9
1GiB
E: 0% (0/2040MiB)
MB/s
Read (MB/s)
Write (MB/s)
SEQ1M Q8T1
6012.55
3635.23
SEQ1M Q1T1
1930.93
790.20
RND4K Q32T1
458.31
452.62
RND4K Q1T1
56.90
189.75



Use Case	Best File System	Reason
Operating System	ext4	Fast random I/O
Windows System	NTFS	Feature-rich
Media Storage	NTFS / ext4	Good sequential read
USB Drives	FAT32	Compatibility
Databases / Apps	ext4	High random performance

Table (in MB/s):

READ	ext2	ext3	ext4	NTFS	FAT32
SEQ1MQ8T1	5246.81	5617.68	5688.66	6182.82	6012.55
SEQ1MQ1T1	2222.73	2395.4	2377.19	3038.04	1930.93
RND4KQ32T1	1859.69	1840.78	1872.59	560.07	458.31
RND4KQ1T1	83.5	83.88	84.06	63.71	56.9
WRITE					

SEQ1MQ8T1	**2627.72**	**3270.77**	**3873.76**	**1086.55**	**3635.23**
SEQ1MQ1T1	**2165.39**	**2527.45**	**2327.13**	**570.7**	**790.2**
RND4KQ32T1	**1303.93**	**1225.42**	**1261.27**	**487.94**	**452.62**
RND4KQ1T1	**429.55**	**418.36**	**428.48**	**264.27**	**189.75**

Conclusion : The benchmarking study clearly shows that ext4 is the best-performing and most efficient file system overall due to Extent-based allocation, Optimized journaling and Superior random and write performance. NTFS is highly optimized for sequential read operations, making it suitable for Windows environments and large file handling.